Week 3 — Building Footprint Data Loading and Integration Report

End-to-end record of how the Microsoft and Google Open Buildings datasets were downloaded, filtered to PDET territory, loaded into MongoDB, and audited. Reproducible with the commands in §6.

1. Sources

Dataset	Provider	Tiles	Raw size
Microsoft Global Building Footprints	Bing / Maxar / Airbus	232 quadkey tiles (Colombia)	~647 MB compressed
Google Open Buildings v3	Google Research	8 S2 level-4 cells	~2.9 GB compressed

Both datasets are openly licensed (ODbL) and cover Latin America. Full provenance and download dates are in [data/raw/SOURCES.md](../data/raw/SOURCES.md).

2. Data acquisition & verification

2.1 Download

scripts/download_buildings.py fetches both datasets:

```
python scripts/download_buildings.py         # both sources
python scripts/download_buildings.py --ms     # Microsoft only
python scripts/download_buildings.py --google # Google only
```

Microsoft: the script fetches the dataset-links index from Azure Blob Storage, filters to Location=Colombia (232 tiles, ~647 MB), and downloads in parallel (8 threads). Each tile is a .csv.gz of newline-delimited GeoJSON features.

Google: the script uses the score_thresholds_s2_level_4.csv index to identify the 8 S2 cells that intersect the PDET bounding box (lon_min=-79.01, lat_min=-0.71, lon_max=-70.00, lat_max=11.35). Tiles are .csv.gz files with columns: latitude, longitude, area_in_meters, confidence, geometry (WKT), full_plus_code.

Downloads are resumable: a tile whose local file matches the expected size is skipped.

2.2 Spatial filtering strategy

The spatial join is performed at insert time (not as a separate tagging pass):

1. The 169 PDET municipality polygons are pulled from upme.municipalities and indexed in a Shapely STRtree.
2. For each feature, the polygon's representative_point() (interior point) is tested against the STRtree. Features whose centroid falls outside every PDET polygon are dropped.
3. Only in-PDET buildings are inserted, with divipola already set.

This eliminates a costly post-load update step and keeps the collections clean.

3. Data integrity & format

3.1 Collections & schema

Both datasets are stored in separate MongoDB collections (buildings_ms, buildings_google) sharing the same $jsonSchema validator (schema/buildings.schema.json). Required fields per document:

Field	Type	Notes
source	string	"microsoft" or "google"
source_id	string	Unique tile-level identifier
divipola	string	5-digit PDET municipality code
area_sqm	number	≥ 0, m² in EPSG:9377 (MS) or from Google metadata
geometry	GeoJSON Polygon	WGS84
loaded_at	string	ISO-8601 UTC

confidence is optional (MS uses −1 as sentinel → stored as null; Google provides values in [0.5, 1.0]).

3.2 Area computation

- Microsoft: polygons are reprojected to EPSG:9377 (MAGNA-SIRGAS Origen

Nacional, the equal-area CRS for Colombia) using pyproj.Transformer and Shapely before area is computed.

- Google: area_in_meters is taken directly from the source CSV

(pre-computed by Google). The EPSG:9377 fallback is used when that field is missing or non-numeric.

3.3 Spatial index

Each buildings collection has a 2dsphere index on geometry, enabling $geoWithin and $geoIntersects queries for the Week 4 analysis. A compound index on (source, divipola) supports per-municipality aggregations.

3.4 Geometry validity issues

A small number of polygons in both datasets fail MongoDB's 2dsphere geo-key extraction (self-intersecting or otherwise degenerate geometries). These are surfaced in the load log and skipped via insert_many(ordered=False). They represent < 0.01 % of the inserted corpus and have no material effect on results.

3.5 Known limitation — San José de Uré (DIVIPOLA 23580)

GADM 4.1 (the geometry source used for the Week 2 municipality re-load on this machine) does not include San José de Uré, a municipality created in 2012 by division of Montelíbano. The municipalities collection therefore contains 169 of the 170 PDET polygons. Buildings in the San José de Uré territory are absent from both building collections; this affects < 0.05 % of the PDET building stock. The original Week 2 run used DANE MGN 2025 and did include this municipality.

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4. Data loading efficiency

Source	Tiles	Rows seen	Inserted	Off-PDET dropped	Load time	Throughput
Microsoft	232	8,221,407	1,763,356	6,457,876	4.5 min	~6,500 ins/s avg
Google	8	31,499,115	2,691,812	28,807,164	9.1 min	~4,900 ins/s avg

The high off-PDET drop rate (~79 % MS, ~91 % Google) is expected: the tiles cover entire S2 cells or quadkeys, most of which extend well beyond the 170 PDET municipalities.

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5. Initial data audit (EDA)

Full EDA output: [docs/week3-eda.md](week3-eda.md)

5.1 Totals

Source	Documents	Total rooftop area (km²)	Mean area (m²)
Microsoft	1,763,356	221.18	125.43
Google	2,691,812	222.90	82.81

Total rooftop area is nearly identical across sources (~222 km²), which gives confidence in the spatial filtering. Google reports 53 % more buildings than Microsoft for the same territory, reflecting denser detection of small structures (mean area 82.8 m² vs. 125.4 m² for MS).

5.2 Cross-source coverage

- Munis with MS detections: 169 / 170
- Munis with Google detections: 165 / 170
- Munis covered by both: 165
- 4 munis covered by MS only, 0 by Google only
- 126 of 165 shared munis have building counts differing > 20 % between sources

The large divergence in per-municipality counts (~76 % of shared munis) underlines why comparing both datasets is required by the project mandate.

5.3 Top municipalities by rooftop area (MS)

Santa Marta (14.2 km²) and Valledupar (13.5 km²) lead, consistent with their status as large departmental capitals within PDET territory. Buenaventura, Florencia, and Turbo follow.

5.4 Google confidence distribution

All Google buildings inserted have `confidence ≥ 0.6` (the default threshold in Google's v3 release). Distribution:

Confidence bucket	Count
[0.6, 0.7)	429,902
[0.7, 0.8)	1,142,547
[0.8, 0.9)	1,042,360
[0.9, 1.0]	77,003

~42 % of Google detections have confidence ≥ 0.8. The Week 4 analysis will report results both inclusive and filtered to `confidence ≥ 0.7` to assess sensitivity.

5.5 Building area distribution (m²)

Source	p10	p50	p90	p99	max
Microsoft	28.85	76.99	229.46	908.33	44,667
Google	16.72	58.05	160.30	447.13	42,207

Google detects substantially smaller footprints at every percentile, which explains the higher document count at similar total area.

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6. Reproducible commands

```
# 0. Prerequisites: MongoDB running, venv with requirements installed
python -m venv .venv && .venv/bin/pip install pymongo shapely pyproj requests geopandas pandas fo...

# 1. Initialize collections, validators, indexes (idempotent)
sed 's|/schema/|'"$(pwd)"'/schema/|g' scripts/init.js | mongosh "mongodb://localhost:27017/" --quiet

# 2. Build and load PDET municipalities (Week 2 re-run on local MongoDB)
#    Requires data/raw/municipios_colombia.geojson (DANE MGN 2025, not committed).
#    Alternative: use GADM 4.1 via the build_pdet_geojson.py helper (see SOURCES.md).
MONGO_URI="mongodb://localhost:27017/" python scripts/load_municipalities.py

# 3. Download building tiles (resumable)
MONGO_URI="mongodb://localhost:27017/" python scripts/download_buildings.py

# 4. Load buildings into MongoDB
MONGO_URI="mongodb://localhost:27017/" python scripts/load_buildings.py

# 5. Generate EDA report
MONGO_URI="mongodb://localhost:27017/" python scripts/eda_buildings.py
# → docs/week3-eda.md
```

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7. MongoDB state after loading

```
upme.municipalities:   169 documents  (2dsphere + unique divipola indexes)
upme.buildings_ms:   1,763,356 documents  (2dsphere + divipola + source indexes)
upme.buildings_google: 2,691,812 documents  (2dsphere + divipola + source indexes)
upme.results:           0 documents  (populated in Week 4)
```